Cloud Computing

Training / Learning Documentation

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What is Cloud Computing?

Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet (“the cloud”) instead of running them on your own physical machines.

- **Servers** provide the processing power to run applications and handle workloads, but instead of buying and maintaining hardware, you rent virtual servers that scale up or down as needed.
- **Storage** offers secure, flexible space to keep files, backups, and large datasets, accessible anytime without worrying about local disk limits.
- **Databases** organize and manage structured information (like customer records or transactions) so applications can query and update data efficiently.
- **Networking** ensures that all these resources communicate seamlessly, connecting users, applications, and services securely across the globe.
- **Software** is delivered as ready-to-use applications or platforms (like email, CRM, or analytics tools), removing the need for installation, updates, or maintenance on local machines.

By delivering these services online, cloud computing eliminates the cost and complexity of owning physical infrastructure, while giving businesses and individuals **scalability, flexibility, and global accessibility**.

Instead of buying and maintaining hardware, you **rent resources on-demand** from providers like Amazon Web Services, Microsoft Azure, or Google Cloud Platform.

Why do we need Cloud Computing?

Without Cloud (Traditional Approach)

- You must **buy servers and hardware**
- High **initial investment**

- Need IT team for **maintenance & updates**
- Limited scalability (hard to handle sudden traffic)
- Risk of downtime if systems fail

With Cloud

- No need to buy hardware → **pay-as-you-use**
- Easily **scale up or down** (auto-scaling)
- High **availability & backup systems**
- Faster deployment (minutes instead of weeks)
- Managed services (databases, AI tools, etc.)

Small Real-World Example

Imagine you're building an **online shopping website**:

Without Cloud

- You buy servers and set them up
- During a festival sale, traffic spikes
  → Your server crashes
  → Customers leave

With Cloud

- You host your app on cloud platforms
- During sale, cloud automatically **adds more servers**
- After sale, resources scale down
  → You only pay for what you used

Advantages of Cloud Computing

1. Cost Efficiency

Cloud computing eliminates the need for purchasing and maintaining physical hardware. Organizations follow a pay-as-you-use model, which reduces capital expenditure.

2. Scalability and Flexibility

Resources can be scaled up or down based on demand. This is useful for handling varying workloads without over-provisioning.

3. High Performance

Cloud providers such as Amazon Web Services, Microsoft Azure, and Google Cloud Platform offer reliable and high-speed infrastructure with minimal latency.

4. Accessibility

Applications and data can be accessed from anywhere with an internet connection, enabling remote work and global collaboration.

5. Backup and Disaster Recovery

Cloud platforms provide automated backup and recovery solutions, reducing the risk of data loss.

6. Reduced Maintenance

Cloud providers manage hardware, updates, and system maintenance, allowing organizations to focus on core activities.

7. Faster Deployment

Services and applications can be deployed quickly, improving time-to-market.

8. Security

Cloud providers implement strong security measures such as encryption, authentication, and monitoring systems.

9. Improved Collaboration

Multiple users can access and work on shared data in real time, improving team productivity.

What is Microsoft Azure?

Microsoft Azure is a **cloud computing platform** created by Microsoft.

Microsoft Azure = A platform that gives you access to:

- Servers (virtual computers)
- Storage (save files online)
- Databases (organized data storage)
- Networking (connect systems)
- Software tools

Key Azure Services

1. Virtual Machines (VM)

A **virtual machine** is a computer inside a computer.

- You don't buy a real PC
- Azure gives you one online

You can install software, run apps, etc.

2. Storage

Azure lets you store:

- Files
- Images
- Videos
- Backups

Think of it like a very large Google Drive, but for companies.

3. Database

A **database** is a structured way to store data.

Example:

- Student records
- Customer details

Azure provides ready-made databases so you don't have to build one.

4. Networking

Networking connects systems together.

Azure helps:

- Connect apps to users
- Connect servers to each other

5. App Services

This lets you **host websites and apps** without managing servers.

5 real-time use cases of cloud computing

1. Online Storage and Backup

Example: Google Drive, Dropbox

- When you upload files, they are stored on cloud servers
- You can access them from any device

Real-time use: Saving notes, photos, or project files

2. Streaming Services

Example: Netflix, Spotify

- Movies and songs are not stored on your phone
- They are streamed from cloud servers

Real-time use: Watching movies or listening to music instantly

3. Social Media Platforms

Example: Instagram, Facebook

- Photos, videos, and messages are stored in the cloud
- Millions of users access data at the same time

Real-time use: Uploading a photo and friends seeing it instantly

4. Online Meetings and Collaboration

Example: Zoom, Microsoft Teams

- Meetings run on cloud servers
- Data is shared in real time

Real-time use: Attending online classes or team meetings

5. E-commerce Websites

Example: Amazon, Flipkart

- Product data, payments, and user traffic handled by cloud
- During sales, cloud scales automatically

Real-time use: Ordering products during big sales without site crashing